Research Methodology

Week 7 Assignment: Questions, Answers and Explanations

Week 7: Design of Experiments

NPTEL course noc26_ge79 (SWAYAM), 10 questions, 1 point each

The portal records no submission from you for this week, so it shows 0 on every question. The correct answers below are the portal's accepted answers. For every option the table gives a verdict and the reason, and each question ends with a short explanation of the correct answer.

Answer key at a glance

Question	Type	Correct option(s)	Portal score
1	Single answer	C	Not submitted
2	Single answer	C	Not submitted
3	Select all that apply	A, D	Not submitted
4	Single answer	C	Not submitted
5	Single answer	B	Not submitted
6	Single answer	D	Not submitted
7	Single answer	A	Not submitted
8	Single answer	D	Not submitted
9	Single answer	D	Not submitted
10	Single answer	B	Not submitted

Questions with explanations

Question 1 of 10

In the true experimental research design:

Single answer.

Option	Statement	Verdict	Reason
A	No control on external variables	**Wrong**	A true experiment is defined by controlling outside factors.
B	Less control on external variables	**Wrong**	Less control describes weaker (quasi-experimental) designs.
C	**Control on external variables by randomization**	**Correct**	Random assignment spreads outside factors equally across groups.
D	No use of randomization	**Wrong**	Randomization is a defining feature of a true experiment.

Correct answer: C.

Explanation: In a true experiment subjects are randomly assigned to groups, so extraneous variables affect all groups alike and their effect is controlled.

Question 2 of 10

Which of the following is an incomplete experimental research design?

Single answer.

Option	Statement	Verdict	Reason
A	Before and after with control group design	**Wrong**	It has a control group for comparison.
B	Factorial design	**Wrong**	It is a complete design that studies several factors together.
C	**One group before-after design**	**Correct**	It has no control group and no randomization, so any change cannot be separated from other influences.
D	Latin Square design	**Wrong**	It is a complete, balanced design.

Correct answer: C.

Explanation: A design with a single group measured before and after the treatment cannot show that the treatment caused the change. That is why it is called incomplete (pre-experimental).

Question 3 of 10

Coefficient of Determination is said to be unity when -

Select all that apply.

Option	Statement	Verdict	Reason
A	**Error Sum of Squares is zero**	**Correct**	$R^2 = 1 - SSE/SST$, which equals 1 when SSE = 0.
B	Total Sum of Squares is zero	**Wrong**	If SST = 0 the ratio is undefined, not 1.
C	Regression Sum of Squares is zero	**Wrong**	SSR = 0 gives $R^2 = 0$.
D	**Regression Sum of Squares is same as Total Sum of Squares**	**Correct**	$R^2 = SSR/SST$ equals 1 when SSR = SST.

Correct answer: A and D.

Explanation: The coefficient of determination is $R^2 = SSR/SST = 1 - SSE/SST$, where SSR is the regression (explained) sum of squares, SST the total and SSE the error (unexplained) sum of squares. $R^2 = 1$ means the model explains all the variation: SSR = SST, which is the same as SSE = 0.

Question 4 of 10

The probability density function describes the following -

Single answer.

Option	Statement	Verdict	Reason
A	Probability of a phenomenon occurring in macro scale	**Wrong**	Scale has nothing to do with the definition.
B	Probability of a phenomenon occurring in micro scale	**Wrong**	Scale has nothing to do with the definition.
C	**Distribution of probabilities in the continuous random variable domain**	**Correct**	It shows how probability is spread over the values of a continuous variable.
D	Distribution of probabilities in some particular set of undefined variables	**Wrong**	The variables are not undefined; the function is for a continuous random variable.

Correct answer: C.

Explanation: For a continuous variable the probability of any single exact value is zero. The density function describes how probability is distributed over the range, and the area under it between two values gives the probability of falling in that range.

Question 5 of 10

During Response Surface Methodology's Canonical Analysis involving two factors, maximum response is said to be achieved when the eigen values of the B matrix are

Single answer.

Option	Statement	Verdict	Reason
A	all positive	**Wrong**	The surface curves upward in every direction, so the stationary point is a minimum.
B	**all negative**	**Correct**	The surface curves downward in every direction, so the stationary point is a maximum.
C	all zero	**Wrong**	The surface is flat (a ridge), with no single maximum.
D	such that one is positive and the other is negative	**Wrong**	This gives a saddle point, neither maximum nor minimum.

Correct answer: B.

Explanation: The fitted second-order model is $\hat{y} = b_0 + x^T b + x^T B x$, where B holds the squared and cross-product coefficients. The eigenvalues of B give the curvature along the principal directions. All negative eigenvalues mean the response falls away in every direction, so the stationary point is a maximum.

Question 6 of 10

Experiments are conducted in laboratories to

Single answer.

Option	Statement	Verdict	Reason
A	Verify scientific predictions	**Wrong**	True, but only one of the purposes.
B	Obtain trends when theory is not known	**Wrong**	True, but only one of the purposes.
C	Find effect of different variables on the desired response	**Wrong**	True, but only one of the purposes.
D	**All of the above**	**Correct**	Each of the three is a real purpose of experiments.

Correct answer: D.

Explanation: Experiments test theory, reveal patterns where no theory exists and measure how each variable affects the response, so all of the above.

Question 7 of 10

SPC and Design of Experiments are very closely interrelated.

Single answer.

Option	Statement	Verdict	Reason
A	**True**	**Correct**	Statistical process control monitors a process, and design of experiments finds the settings that make it perform best; each feeds the other.
B	False	**Wrong**	Both use statistical methods on process data and support each other.

Correct answer: A.

Explanation: Statistical process control (SPC) keeps a process stable, while design of experiments identifies which factors and settings improve it. Results of one are used in the other, so they are closely related.

Question 8 of 10

Which of these is not an application of experimental design in the field of Engineering design?

Single answer.

Option	Statement	Verdict	Reason
A	Evaluation and comparison of basic design configurations	**Wrong**	This is an application of experimental design.
B	Evaluation of material alternatives	**Wrong**	This is an application of experimental design.
C	Evaluation of key product design parameters that impact performance	**Wrong**	This is an application of experimental design.
D	**Evaluation of correct process to develop the product**	**Correct**	Choosing the manufacturing process is a process-design matter that the key treats as outside engineering design.

Correct answer: D.

Explanation: Experimental design is used in engineering design to compare design configurations, compare materials and find the product parameters that most affect performance. The answer key does not count selecting the process to develop the product as one of these.

Note: *This question asks for what is NOT an application, so "Correct" marks the option you should select. The reasoning for option D is the weakest in the set: the answer key treats choosing the manufacturing process as outside engineering design, so remember it as the key's choice.*

Question 9 of 10

The average estimated from a data collection is 50 cm. The quantity (quantities) which has the same units as the data is (are)

Single answer.

Option	Statement	Verdict	Reason
A	Standard deviation	**Wrong**	True, but not the only quantity in cm.
B	Median	**Wrong**	True, but not the only quantity in cm.
C	Mode	**Wrong**	True, but not the only quantity in cm.
D	**All of the above**	**Correct**	All three are measured in cm.

Correct answer: D.

Explanation: The median and mode are values picked from the data, so they are in cm. The standard deviation is the square root of the average squared deviation, so it returns to cm. (The variance would be in cm^2.)

Question 10 of 10

Sample collected from an instrument giving binary data has the following values (1, −1, −1, 1, 1, −1). Mean of the above data set is

Single answer.

Option	Statement	Verdict	Reason
A	-1	**Wrong**	Only half of the values are −1.
B	**0**	**Correct**	The sum is 0 and there are 6 values, so the mean is 0.
C	1	**Wrong**	Only half of the values are +1.
D	3	**Wrong**	3 is the number of +1 values, not the mean.

Correct answer: B.

Explanation: Sum = 1 − 1 − 1 + 1 + 1 − 1 = 0. Mean = sum divided by the number of values = 0 / 6 = 0. There are three values of +1 and three of −1, which cancel.